

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Purchases

Transactional records

18

Features

Demographics, behavior & purchase details

37

Missing Values

In Review Rating column

\$59.76

Avg. Purchase

Mean spend per order

Data Preparation in Python

→ Clean & Standardize

Imputed missing Review Ratings; renamed columns to snake_case

→ Feature Engineering

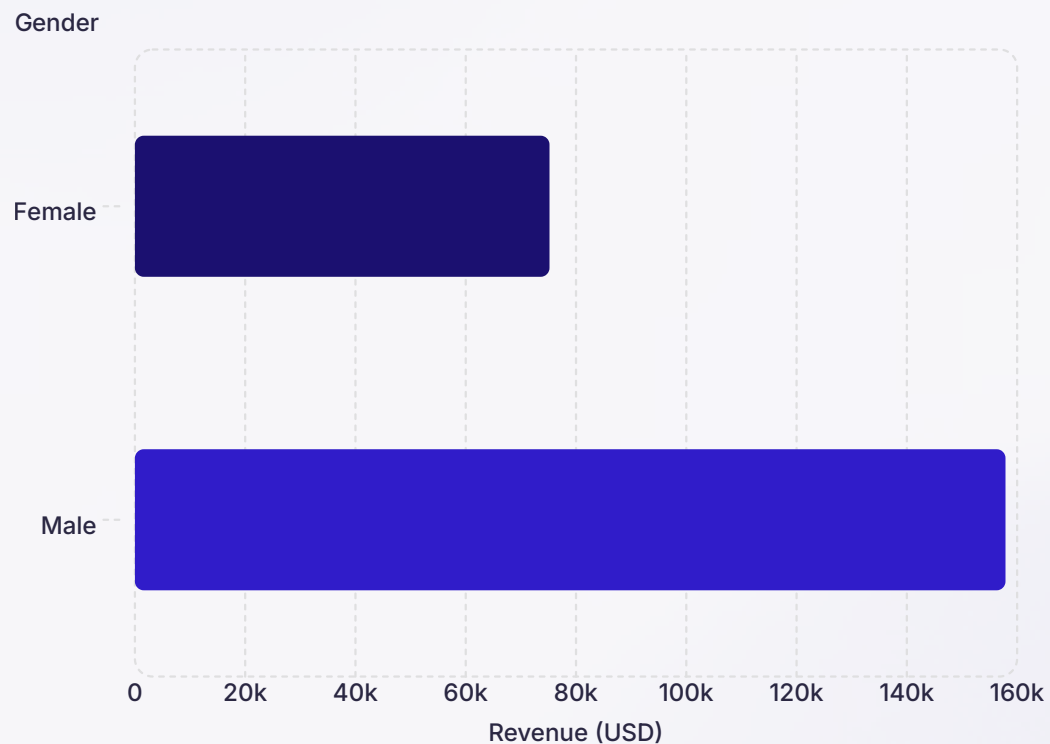
Created `age_group` and `purchase_frequency_days` columns

→ Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for SQL analysis



Revenue by Gender

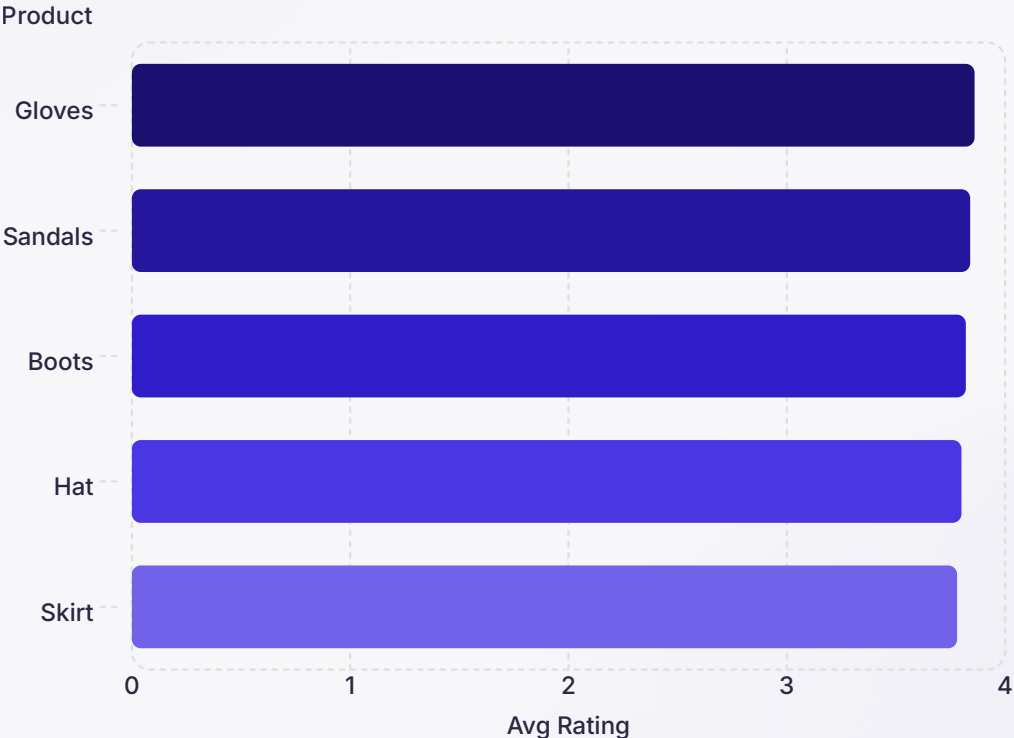


Key Insight

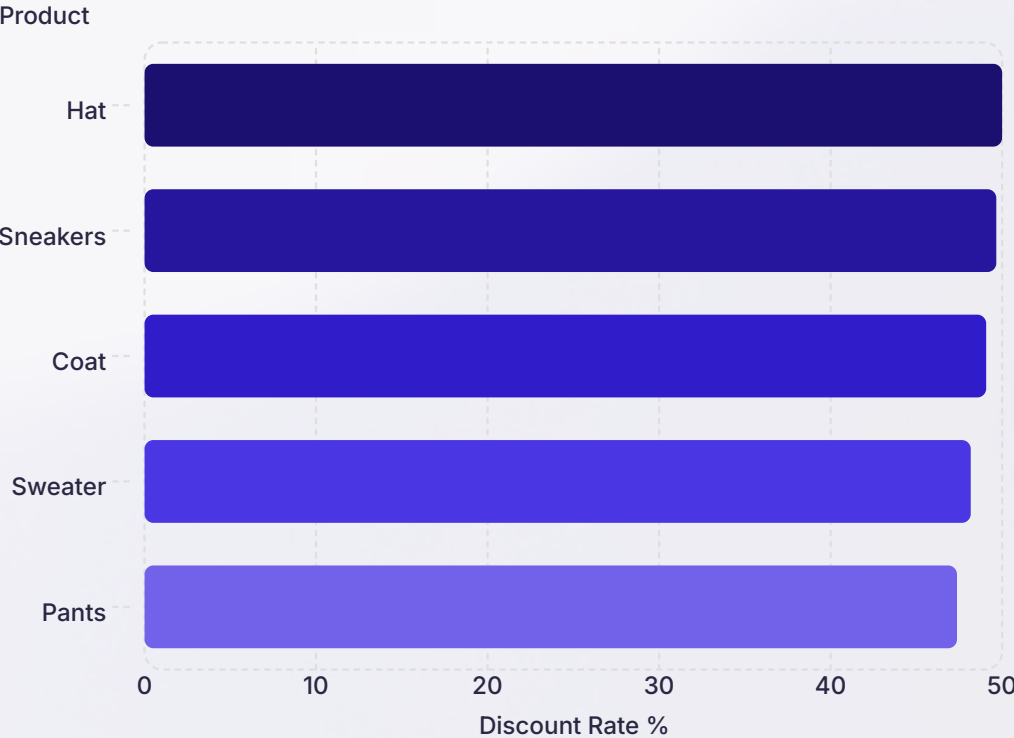
Male customers generated **2.1x more revenue** than female customers — a critical segment for targeted marketing.

Top Products by Rating & Discount Rate

Highest Rated Products

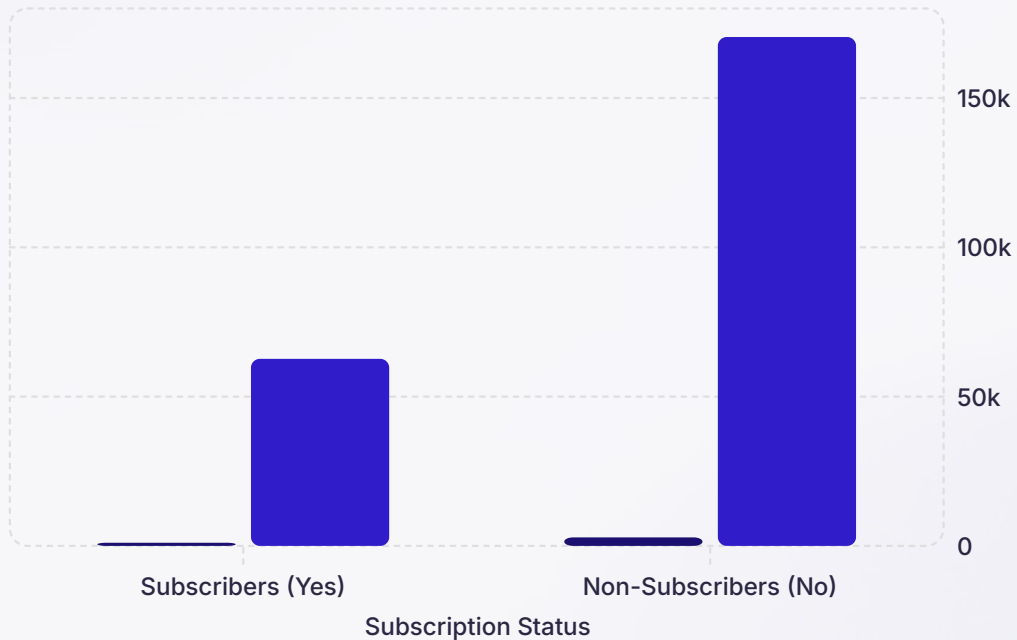


Most Discounted Products



Subscribers vs. Non-Subscribers

■ Customers ■ Revenue (USD)

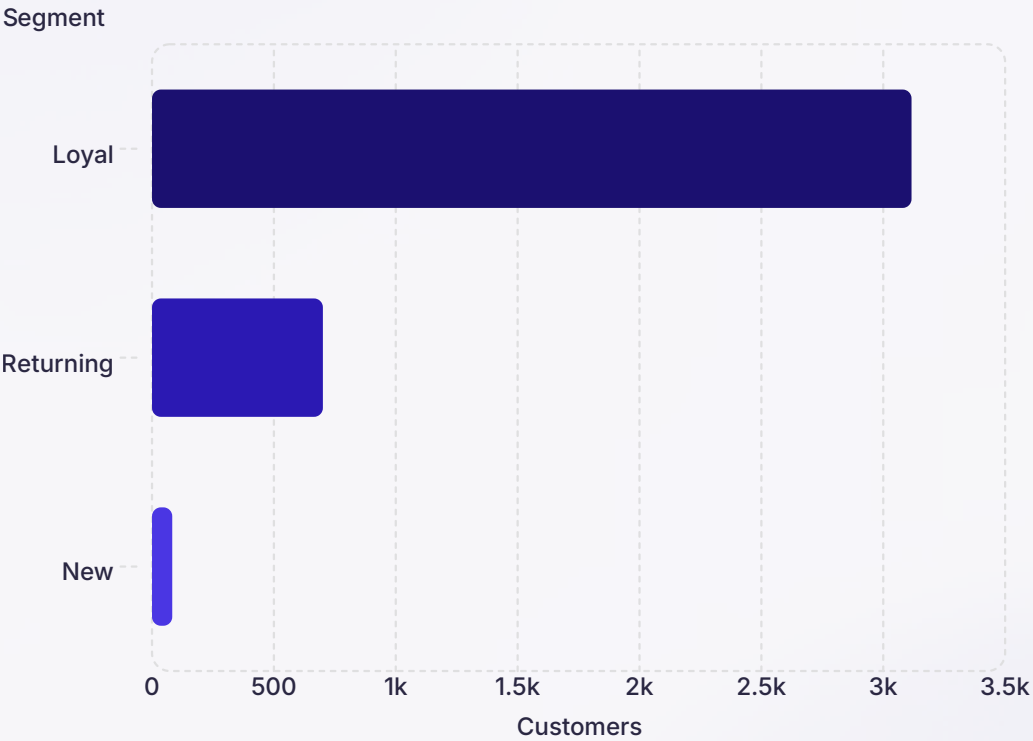


Subscription Gap

Only **27%** of customers subscribe, yet they represent a high-value segment. Non-subscribers generate **73%** of total revenue.

ⓘ Avg spend is nearly identical — subscription is an untapped growth lever.

Customer Segmentation



Segment Breakdown

Loyal customers dominate at 80% of the base. Only **83 New** customers — growth opportunity.

Loyal

3,116 customers

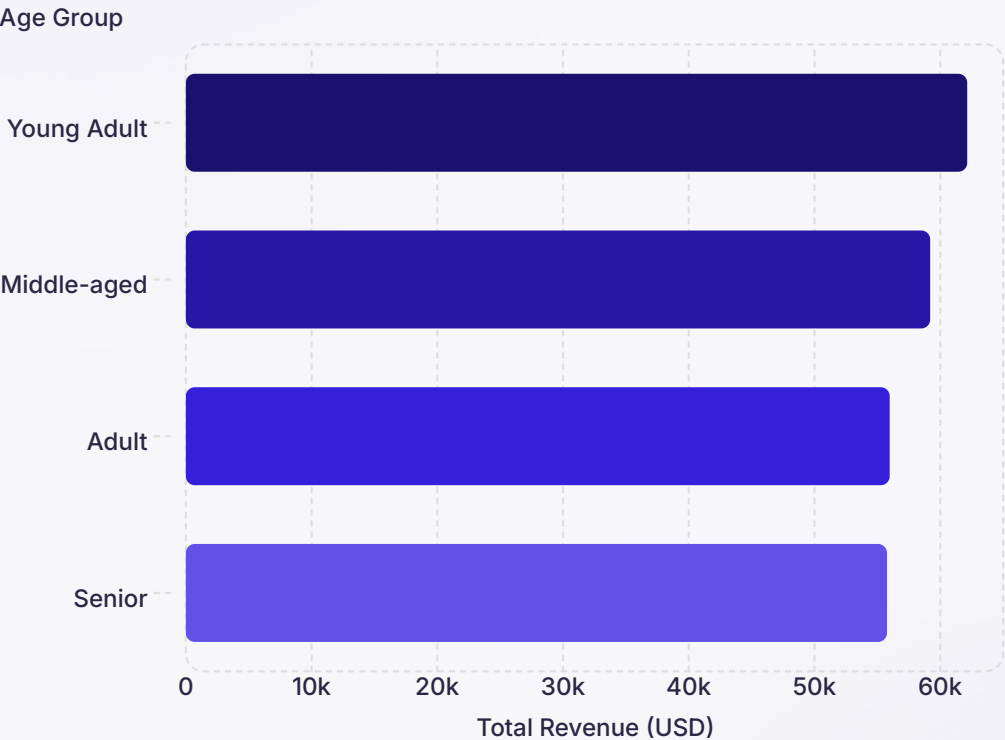
Returning

701 customers

New

83 customers

Revenue by Age Group



Age Group Insights

Young Adults lead in revenue contribution. Revenue is relatively evenly distributed across all age groups.

Express Shipping

Avg purchase \$60.48 vs \$58.46 Standard

Repeat Buyers

958 subscribers among high-frequency purchasers

Power BI Dashboard

3.9K Customers

Total customer base

3.75 Avg Rating

Overall review score

\$59.76 Avg Order

Mean purchase amount

4 Categories

Clothing, Accessories, Footwear, Outerwear





Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users